ECA0806-Analog & Digital Communication

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Experiment-17

PERFORMANCE EVALUATION OF DIGITAL COMMUNICATION SYSTEMS USING EYE DIAGRAM ANALYSIS UNDER ADDITIVE WHITE GAUSSIAN NOISE (AWGN) USING MATLAB

Program (Matlab Code):

```
clc;
clear;
close all;

%% Parameters
N = 100;
sps = 20;
SNR = 10;

%% Generate Random Binary Data
rng(1);
data = randi([0 1], 1, N);

%% NRZ Encoding
tx_signal = repelem(data, sps);

%% Add AWGN Manually
signal_power = mean(tx_signal.^2);
noise_power = signal_power / (10^(SNR/10));

noise = sqrt(noise_power) * randn(size(tx_signal));
rx_signal = tx_signal + noise;

%% Time Axis
t = (0:length(tx_signal)-1)/sps;

%% Figure 1: Manual-Style Four Subplots
figure(1);

subplot(4,1,1);
stairs(0:N-1, data, 'LineWidth', 1.2);
grid on;
title('Original Binary Data');
xlabel('Bit Number');
```

```
ylabel('Bit');
xlim([0 N]);
ylim([-0.2 1.2]);

subplot(4,1,2);
plot(t, tx_signal, 'LineWidth', 1.2);
grid on;
title('Transmitted NRZ Signal');
xlabel('Time');
ylabel('Amplitude');
ylim([-0.5 1.5]);

subplot(4,1,3);
plot(t, rx_signal, 'r');
grid on;
title(['Received Signal with AWGN (SNR = ', ...
      num2str(SNR), ' dB)']);
xlabel('Time');
ylabel('Amplitude');

subplot(4,1,4);
hold on;

% Overlay consecutive two-bit segments
L = 2*sps;
for k = 1:L:length(rx_signal)-L
   segment = rx_signal(k:k+L-1);
   plot((0:L-1)/sps, segment, 'b');
end

grid on;
title('Eye Diagram of Received Signal');
xlabel('Time (bit periods)');
ylabel('Amplitude');
xlim([0 2]);
hold off;

%% Figure 2: Eye Diagrams at Different SNR Values
SNR_values = [20 10 5 0];

figure(2);

for m = 1:length(SNR_values)
```

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```
    noise_power = signal_power / ...
                (10^(SNR_values(m)/10));

    rx_temp = tx_signal + ...
            sqrt(noise_power)*randn(size(tx_signal));

    subplot(2,2,m);
    hold on;

    for k = 1:L:length(rx_temp)-L
        segment = rx_temp(k:k+L-1);
        plot((0:L-1)/sps, segment, 'b');
    end

    grid on;
    title(['Eye Diagram - SNR = ', ...
          num2str(SNR_values(m)), ' dB']);
    xlabel('Time (bit periods)');
    ylabel('Amplitude');
    xlim([0 2]);
    hold off;
end

disp('Both figures generated successfully.');
```

Output Waveform :

